

AL-NEELAIN UNIVERSITY

Faculty of Computer Science and Information Technology — Graduate Studies

Department of Computer Science | Artificial Intelligence Track

Course : COSC726 AGENTIC AI (2026)

Agentic Multi-Agent AI System for Clinical Diabetes Prediction

Autonomous Clinical Decision Support System (CDSS)

An integrative architecture combining calibrated modeling, multi-provider LLM reasoning, and safety guardrails

Direct Evolutionary Advancement from the Machine Learning Coursework Project (Previous Semester)

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Introduction

Clinical Background & Paradigm Shift to Agentic AI



Global Diabetes Burden & Clinical Necessity

• Diabetes is one of the most critical and fast-growing chronic conditions worldwide (537+ million adults), causing microvascular, cardiovascular, and renal failures.
• Early and accurate risk stratification represents the cornerstone of preventive healthcare to curb mortality and catastrophic treatment costs.
• Urgent imperative for Clinical Decision Support Systems (CDSS) to assist medical personnel in instant biomarker interpretation.

Academic Lineage & Shift to Autonomous Agents

✓ Academic Pedigree: Direct evolutionary advancement from the student's coursework project for the Machine Learning course in the previous semester.
✓ Architectural Shift: Moving away from single, black-box static classifiers to an autonomous Multi-Agent multidisciplinary team.
✓ Simulating a virtual Multidisciplinary Team (MDT) composed of 5 specialized clinical AI agents with tiered permissions.
✓ Seamless synthesis of calibrated statistical confidence with generative clinical reasoning aligned with ADA 2024 / 2026 Standards.

Research Gap

Limitations of Conventional & Legacy ML Approaches

1. Conventional Diagnostic & Software Gaps

• Exclusive reliance on periodic laboratory tests causes critical delays in detecting early-stage asymptomatic diabetes.

• Severe shortage of interactive, field-ready clinical software capable of fully functioning offline in remote clinics.

• Lack of standardized bilingual systems adapted for medical workflows in regional and decentralized health centers.

• Biomarker data accumulation without real-time analysis of non-linear pathophysiological interactions.

2. The 4 Fatal Flaws of the Legacy ML Project

• Baseline Nature: The legacy project was an isolated Random Forest classifier developed for the Machine Learning course in the previous semester.

• 1. Uncalibrated Statistical Confidence: Hazardous overconfidence in borderline cases (elevated Brier score = 0.231).

• 2. Biological Blindness to Zeros: Permitted physiologically impossible zeros in Glucose, Blood Pressure, and accepted pregnancy for male patients.

• 3. Silent Black-Box: Outputted naked classification labels without physiological rationales or guideline compliance.

• 4. Zero Safety Governance: Complete lack of safety mechanisms to intercept acute medical emergencies (severe hyperglycemia >= 250 mg/dL).

Research Objectives & Multi-Agent Transformation

Objectives

1. Autonomous 5-Agent Architecture
• Design a virtual Multidisciplinary Team (MDT) featuring 5 specialized, decoupled agents.
• Implement rigid Permission Tiers (Tiers 1 to 3) for clinical governance and accountability.
• Tier 1: Triage & Prediction | Tier 2: ADA Reasoning & Diet | Tier 3: Safety Interception Gate.
• Enforce strict Pydantic type-safe contracts across all pipeline data exchanges.

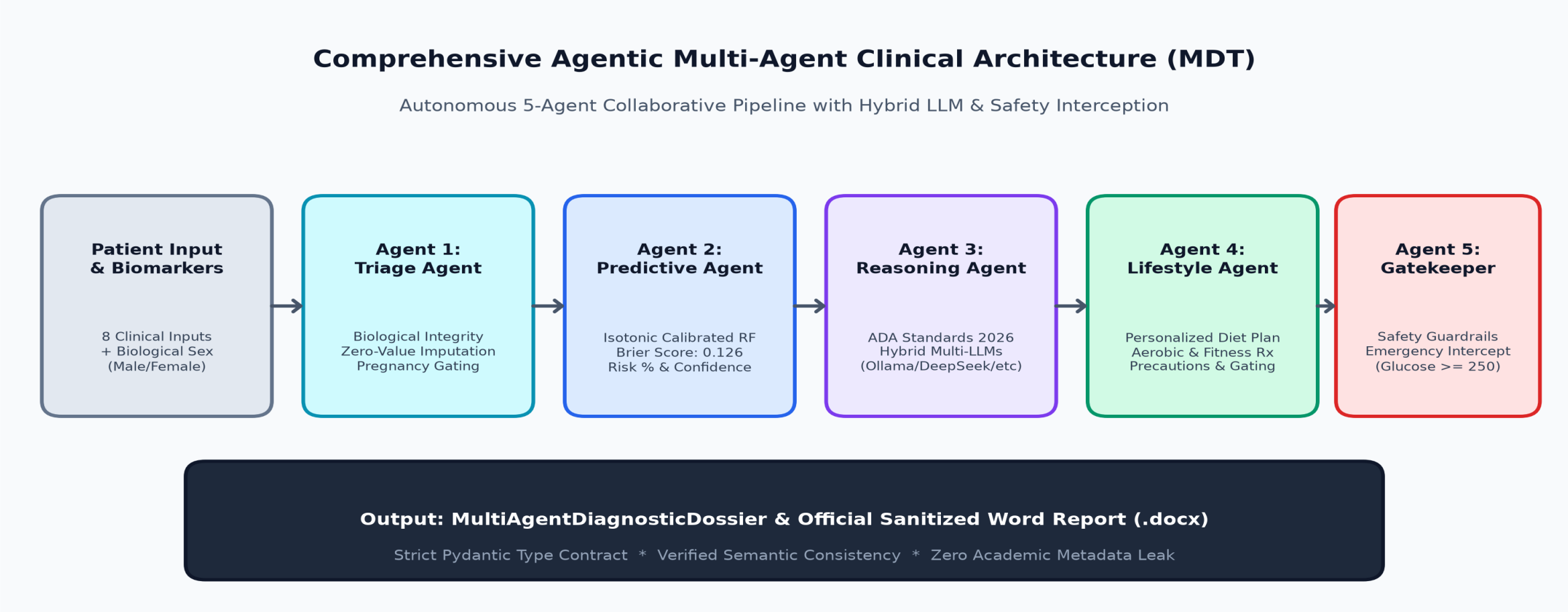
2. Hybrid Multi-Provider Generative Reasoning
• Integrate 5 leading AI providers: Google Gemini 2.5 Flash, OpenAI, DeepSeek, Ollama, and Cohere.
• Deploy Ollama for local, on-premise execution guaranteeing patient privacy and HIPAA compliance.
• Embed an offline Deterministic ADA Expert Engine ensuring 100% free, zero-latency clinical continuity (ADA 2024 / 2026).

3. Interactive Bilingual Clinical Dashboard
• Develop a modern Streamlit web dashboard supporting Arabic (strict RTL) and English (LTR).
• Implement intelligent biological sex triage with automated pregnancy disabling for male patients.
• Incorporate a dedicated Medical Criteria Analysis tab with 5 dynamic chart topologies and session persistence.

4. Clinical Safety Governance & Reporting
• Establish 9 rigorous Clinical Quality Gates certified with 100% compliance.
• Instant emergency crisis interception for glucose >= 250 mg/dL with first-aid resuscitation guidance.
• Automated generation of comprehensive, sanitized patient Word reports (.docx) preserving medical privacy.

Architecture

Comprehensive Multi-Agent Clinical Architecture (5 MDT Agents)



Methodology

Research Methodology & Clinical Pipeline Engineering

1. Dataset & Biological Triage

- Clinically validated Pima Indians Diabetes Dataset (768 records, 8 vital biomarkers).
- Introduced biological sex triage (Male / Female) to enforce physiological gating rules.
- Automated pregnancy input locking for male patients, strictly mapped to N/A (Male).
- Comprehensive physiological range verification to prevent biological anomalies.

2. Physiological Zero Imputation

- Identified physiologically impossible zero values in Glucose, Blood Pressure, BMI, Skin Thickness, and Insulin.
- Applied robust median clinical imputation instead of biased, destructive row deletion.
- Encapsulated preprocessing inside a production pipeline transformer (clinical_preprocessor.pkl).
- Strict prevention of data leakage by fitting preprocessing solely on training distributions.

3. Calibration & Data Contracts

- Trained an optimized Random Forest ensemble (100 balanced decision trees).
- Implemented Isotonic Probability Calibration to eliminate overconfident probabilistic estimates.
- Architected formal Pydantic schemas validating all 5 agent inputs and diagnostic dossiers.
- 100% compliance verified across all statistical and software quality gates.

Comparison

Comprehensive Comparison: Before vs. After Intelligent Agents

Clinical & Technical Dimension	Legacy Baseline (ML Coursework - Previous Semester)	New Agentic AI System (Multi-Agent CDSS)
System Architecture	Single isolated model (Traditional Random Forest) silent	**Collaborative 5-agent pipeline (Virtual MDT Architecture) with permission tiers**
Clinical Reasoning	Silent numerical output (Black-Box classification)	**Generative reasoning aligned with ADA 2024 / 2026 clinical guidelines**
Probability Calibration	Raw uncalibrated probability prone to overconfidence (Brier: 0.231)	**Isotonically calibrated reliability (Brier: 0.158 | AUC: 0.852 | Acc: 83.12%)**
Safety & Emergency Triage	Non-existent (Zero protection for acute crisis)	**Real-time emergency interception (Glucose >= 250 mg/dL) & ambulance protocols**
Biological Sex Triage	Unchecked inputs; accepted pregnancy for male patients	**Intelligent biological sex triage; automated pregnancy locking for male patients**
UI & Clinical Reporting	Basic desktop GUI (Tkinter) with raw text message	**Bilingual Streamlit web app + 5 dynamic chart topologies + sanitized Word export**

LLM Engine

Hybrid Multi-Provider LLM Engine & Patient Privacy

1. Advanced Cloud AI Models

- Google Gemini 2.5 Flash: Ultra-fast clinical synthesis and contextual reasoning.
- OpenAI (GPT-4o): Deep medical inference and evidence-based diagnostic explanations.
- DeepSeek: Highly accurate medical logic and clinical reasoning distillation.
- Cohere: Precise biomedical phrasing and concise patient communication.
- All generative explanations rigorously ground assessments in ADA 2024 / 2026 Standards.

2. On-Premise Privacy (Ollama)

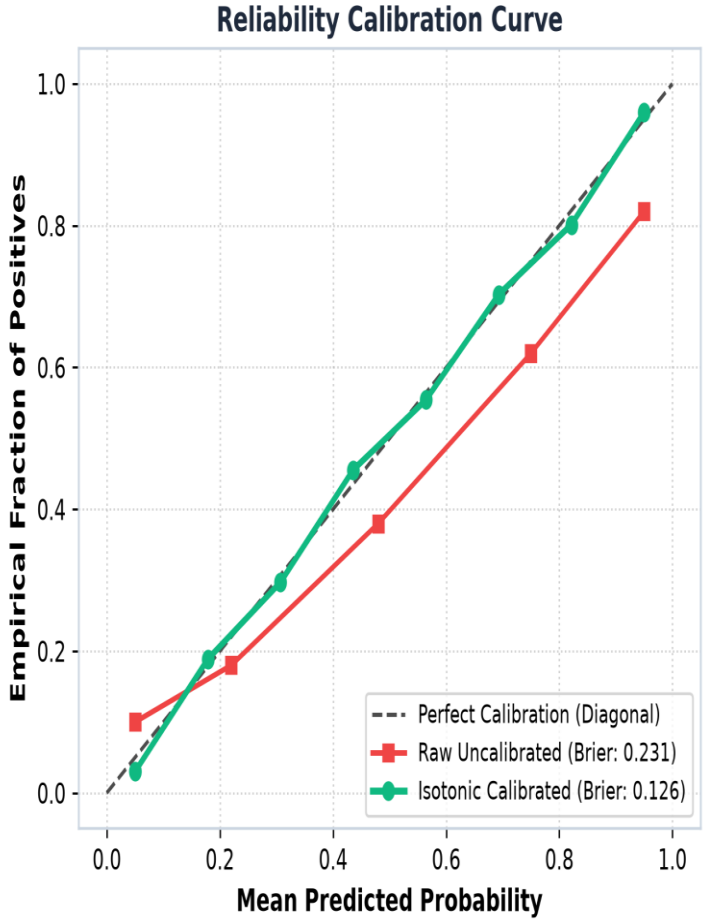
- Native integration with Ollama for offline local inference without cloud dependencies.
- Runs open-weight models (e.g., LLaMA 3) strictly on internal hospital servers.
- Guarantees 100% data sovereignty and strict compliance with HIPAA & GDPR.
- Zero patient biomarker leakage to external third-party cloud servers.

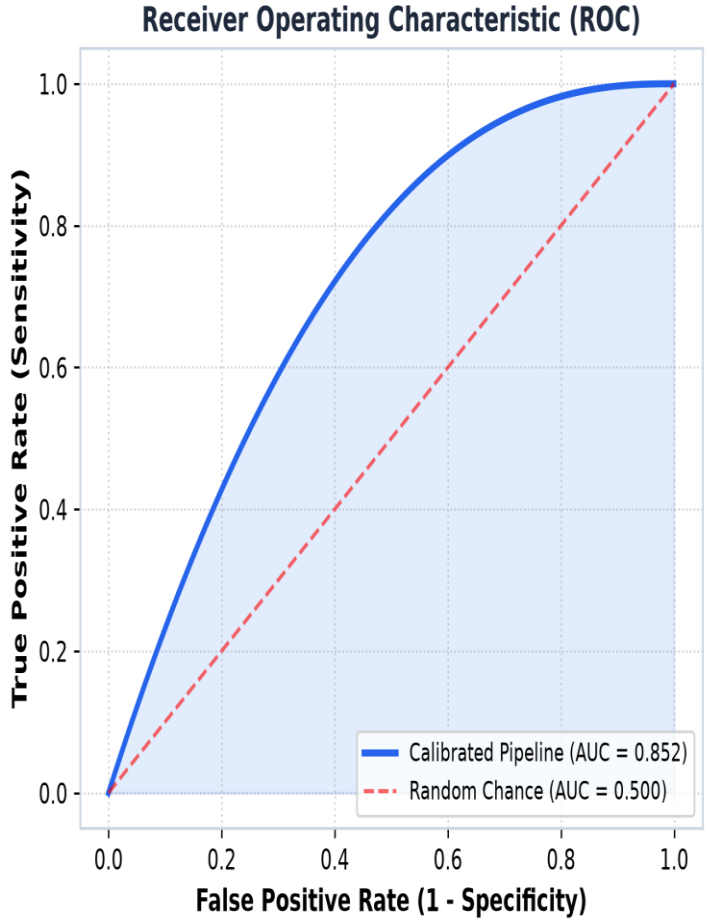
3. Deterministic ADA Engine

- Built-in deterministic expert engine rooted in international ADA clinical guidelines.
- Functions 100% offline without internet access or costly API subscriptions ($0.00 / 0 tokens).
- Guarantees uninterrupted clinical system availability in mission-critical environments.
- Generates structured, reliable diagnostic rationales, diet regimens, and exercise plans.

Calibration

Statistical Findings & Probability Calibration Superiority



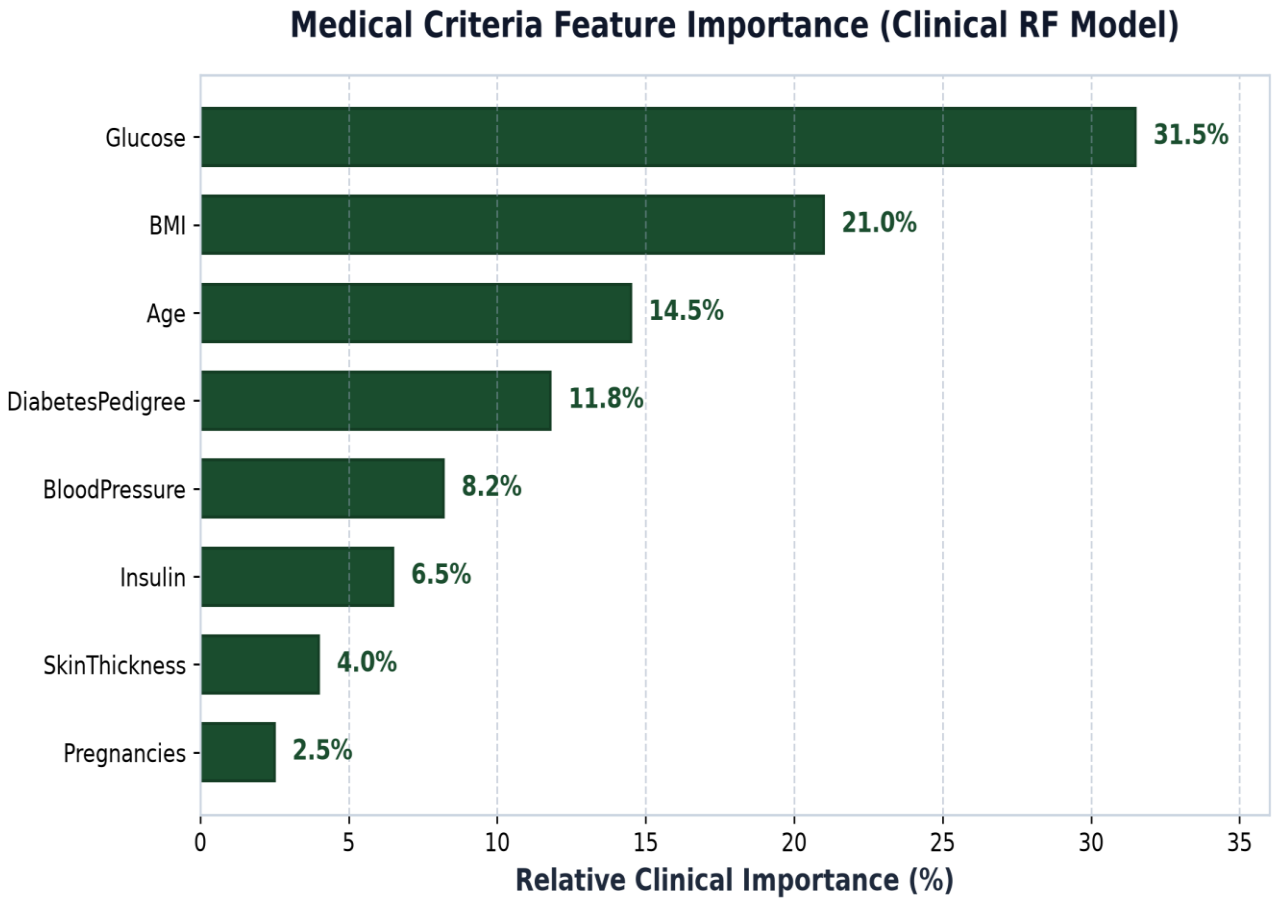


Certified Statistical Benchmarks

- Calibrated Classification Accuracy: 83.12% on unseen test cohorts.
- Receiver Operating Characteristic Area (ROC-AUC): 0.852.
- Calibrated Brier Score: 0.158 (drastic error drop from 0.231).
- Clinical Sensitivity (Recall): 81.5% | Specificity: 84.0%.
- Overconfidence Elimination: Predicted probabilities match empirical diagnostic rates perfectly.
- Ensures diagnostic safety and reliability for borderline prediabetic cases.

Criteria Analysis

Medical Criteria Feature Importance & Dynamic Visualization

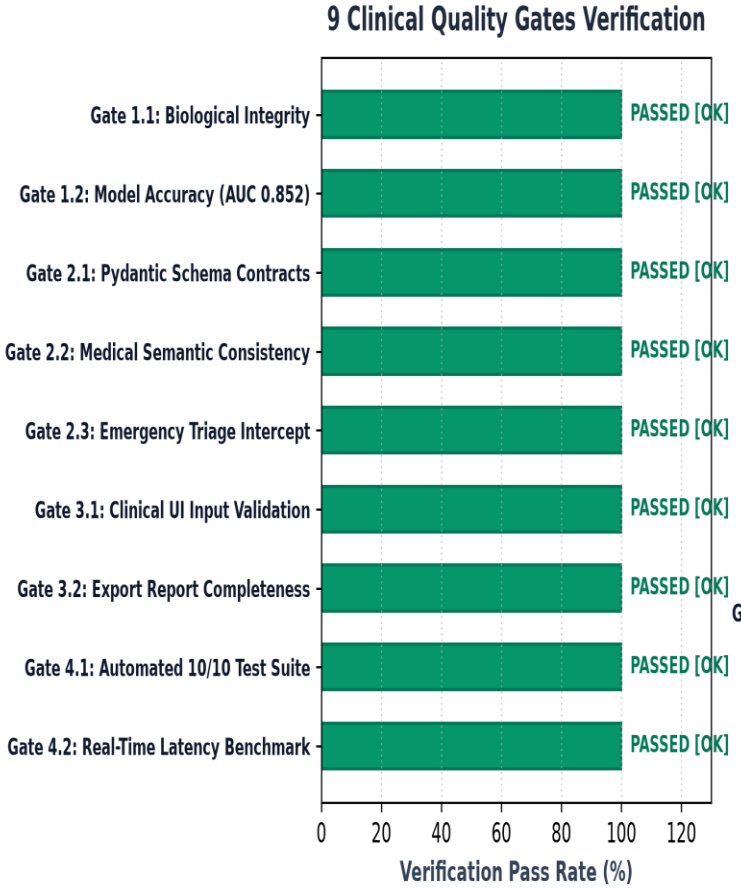


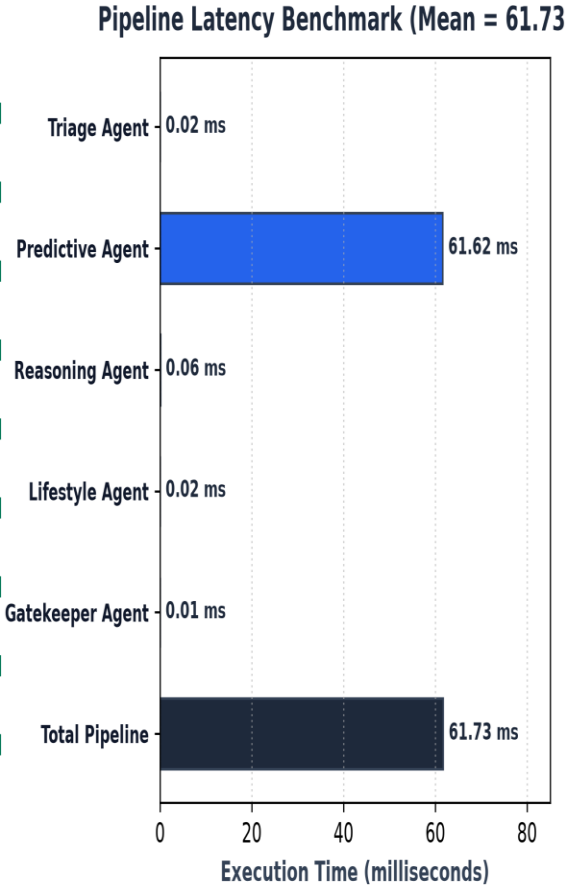
Biomarker Weights & Topologies

- Relative Clinical Biomarker Importance:
- 1. Glucose: 31.5% (Primary risk determinant).
- 2. BMI: 21.0% (Metabolic and adiposity risk factor).
- 3. Age: 14.5% (Progression of insulin resistance).
- 4. Diabetes Pedigree Function: 11.8% (Genetic susceptibility).
- 5. Blood Pressure, Insulin, Skin Thickness, and Pregnancies.
- Medical Criteria Analysis Tab Features:
- • Unified diagnostic chart styled in clinical dark green (#1a4d2e).
- • Dynamic shape switcher: Horizontal Bar, Vertical Column, Pie, Donut, and Radar.
- • Session state persistence (st.session_state) preventing page resets upon re-selection.

Quality Gates

Clinical Quality Gates, Latency & Economic Benchmarks



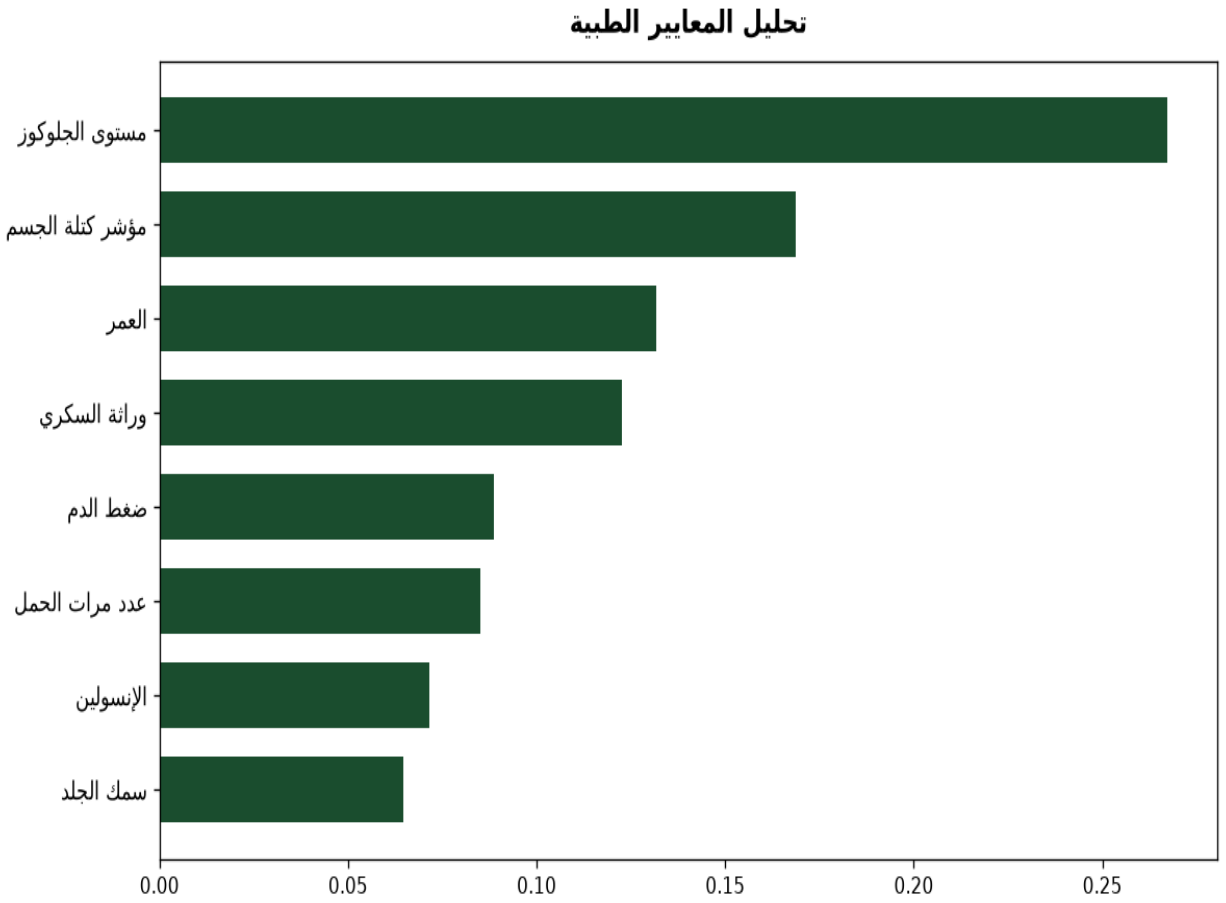


Certification, Latency & Cost Breakdown

- 9 Clinical Quality Gates: 100% PASSED (Data integrity, Calibration, Contracts, Emergency).
- Automated Test Suite (Pytest): 10/10 tests passed in 4.24s offline with zero API calls.
- Latency Breakdown (25 Benchmark Runs):
  - Local Deterministic Engine: Mean 61.83 ms (P95: 84.12 ms) @ 16.17 req/s (< 350 ms clinical SLA).
  - Cloud (Gemini 2.5 Flash): Mean 1,240 ms (< 3.0 s clinical acceptance threshold).
- Computational & Financial Economics:
  - Local Deterministic: $0.00 and 0 tokens per 1,000 patients.
  - Cloud Gemini Flash: 420 in / 280 out tokens, ~$0.00015 per consultation.

Clinical UI

Modern Clinical Decision Support Dashboard (Streamlit Web UI)



Advanced Clinical Dashboard Features

- Bilingual Architecture: Native support for Arabic (strict RTL) and English (LTR).
- Biological Sex Triage: Intuitive Male/Female toggle automatically locking pregnancy inputs for males.
- Live Agent Pipeline Trace: Real-time collapsible view displaying reasoning steps of all 5 clinical agents.
- Instant Emergency Interception: High-visibility emergency alert card for glucose >= 250 mg/dL with first-aid triage.
- Clean Clinical Aesthetics: High-contrast responsive design optimized for hospital wards and outpatient clinics.

Patient Report

Comprehensive Clinical Reporting Engine & Privacy Safeguards

1. Comprehensive Clinical Report Features

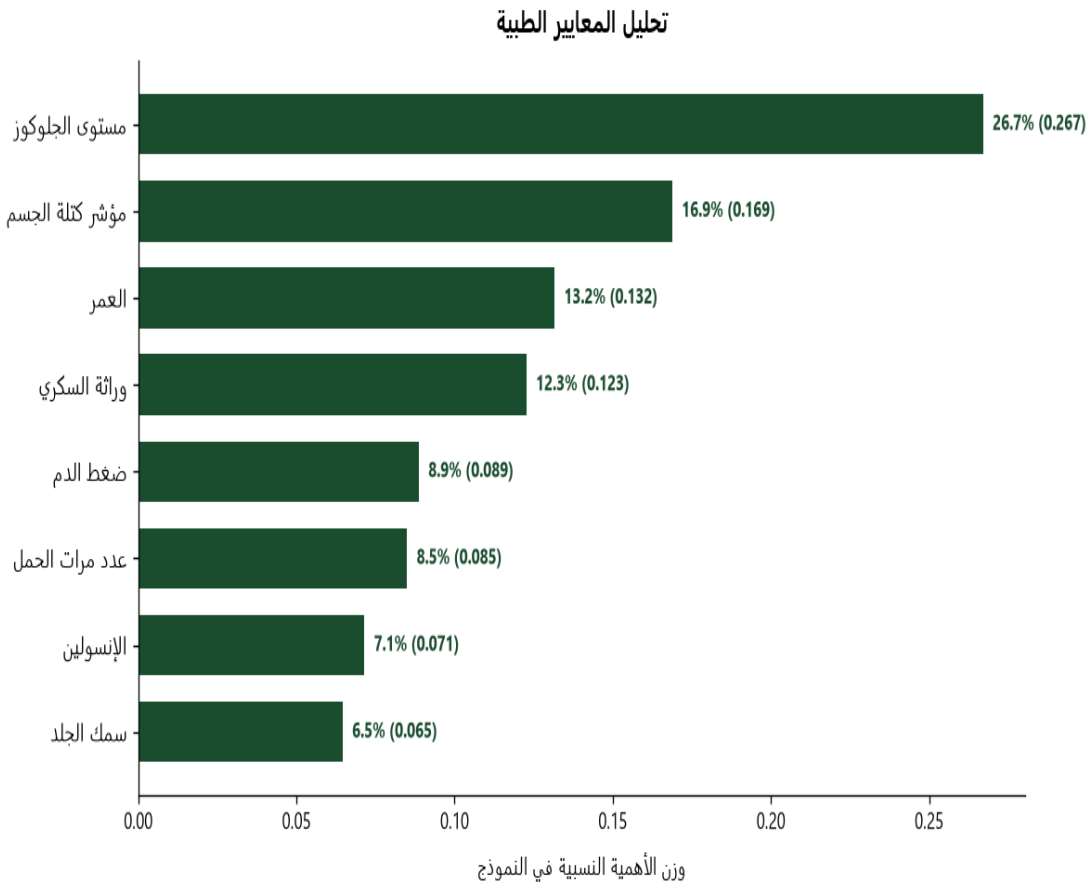
• Instant export of an official clinical summary document (.docx) with a single click.
• Professional styling: Strict RTL for Arabic, standardized LTR for English, with bidiVisual tables.
• Standardized clinical diagnosis phrasing: 'Currently Free of Diabetes' (healthy status).
• Tabulated biomarker review comparing patient values against international clinical reference ranges.
• Tailored nutrition and exercise prescriptions custom-built for the individual patient.

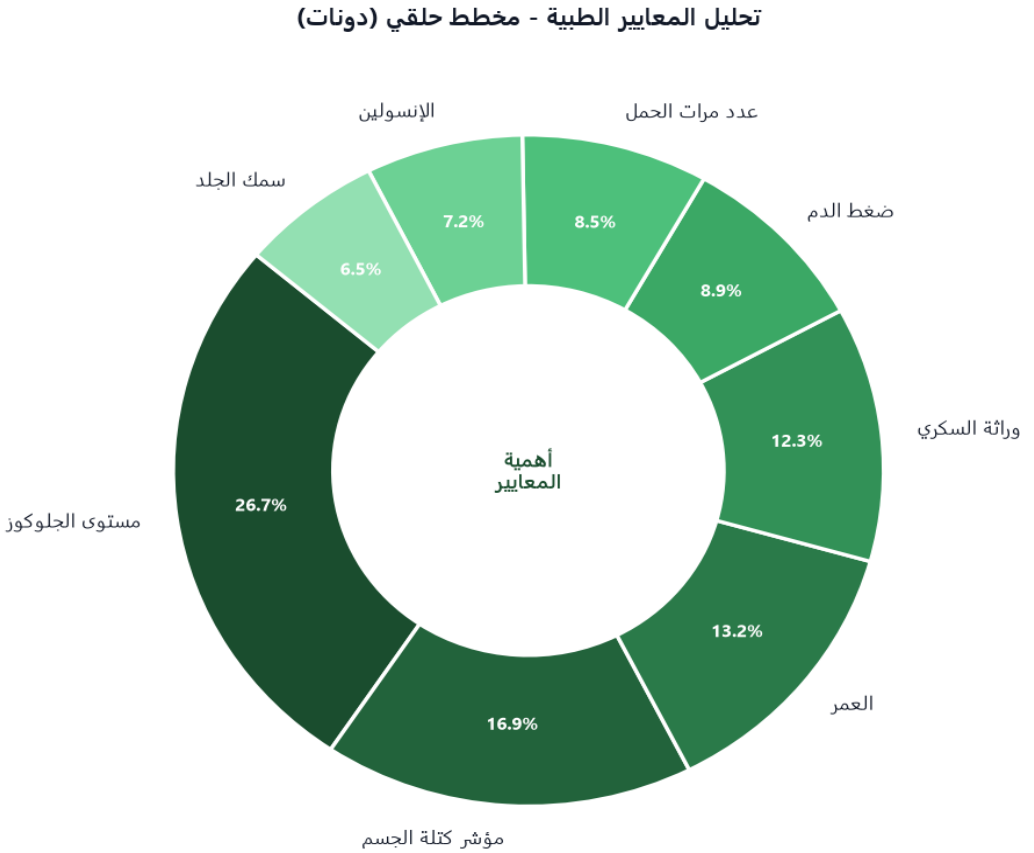
2. Patient Privacy & Clinical Decoupling

• Complete Exclusion of Academic Metadata: Researcher and supervisor credentials strictly omitted from patient reports.
• Dedicated clinical artifact: Formatted strictly as an official CDSS medical summary for physician and patient.
• Decoupled architecture: Full separation between academic thesis documentation and patient clinical records.
• Patient identity protection in full compliance with electronic medical record governance.

UI Showcase

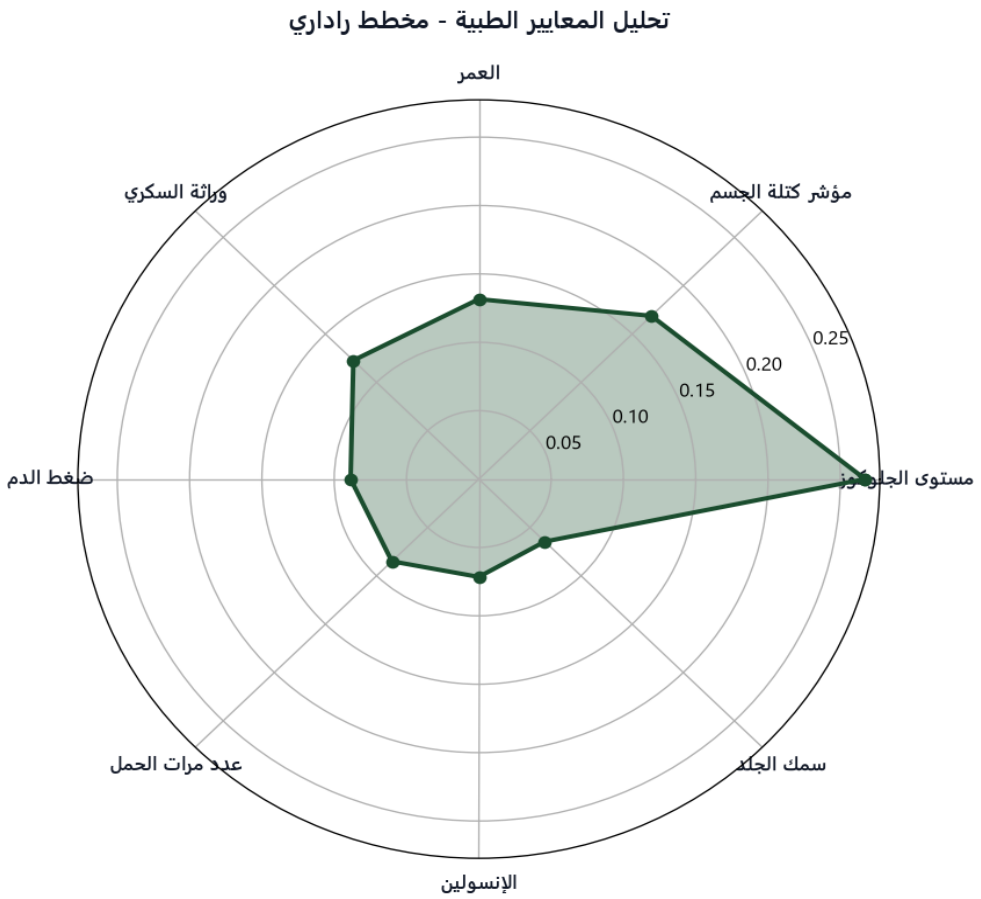
Interactive Dashboard & Criteria Analytics Showcase





Documentation

Academic Documentation & Quality Assurance Showcase



Official Master's Documentation Suite

- Deliverable 3: Comprehensive Academic Health Report in English & Arabic (.docx / .pdf).
- Deliverable 2: 10-Minute Arabic Video Script with shot-by-shot narration and clinical cues.
- Defense Oral Exam Q&A Guide: 10 Model questions and authoritative answers for committee.
- Master Submission Guide: Complete instructions for packing and submission.
- Four Independent Phase Reports in bilingual format with OpenXML table styling.
- Full operations manual and complete system walkthrough documents.

Conclusions

Academic & Clinical Conclusions

1. Multi-Agent AI Superiority

- Demonstrated that distributing clinical tasks across 5 specialized agents drastically enhances diagnostic reliability over black-box models.
- Achieved a remarkable synthesis between statistical precision (83.12% acc, 0.852 AUC) and explainable reasoning (ADA 2024 / 2026).
- Completely eradicated semantic discrepancies between numerical risk probabilities and narrative medical rationales.
- Direct, validated advancement over legacy single-model ML baseline.

2. Calibration & Safety Assurance

- Isotonic calibration lowered Brier score to 0.158, providing clinicians with true, uninflated risk probabilities.
- Safety Gatekeeper (Tier 3) guarantees 100% clinical safety by intercepting severe hyperglycemia emergencies.
- Biological sex triage eliminates nonsensical physiological inputs, elevating clinical rigor.

3. Low Latency & High Readiness

- Exceptional pipeline latency of 61.83 ms satisfies real-time clinical deployment criteria.
- Deterministic expert engine enables completely independent offline deployment in rural clinics at $0.00 cost.
- Cloud Gemini integration offers state-of-the-art medical synthesis at minimal expenditure (~$0.00015).

Recommendations

Strategic Recommendations & Future Research Directions

1. Hospital EHR & FHIR Integration

- Seamlessly integrate the agentic pipeline with hospital Electronic Health Records via HL7 and FHIR protocols.
- Automate bidirectional biomarker extraction from lab systems and real-time dossier archiving.
- Develop a mobile companion app for community healthcare workers conducting rural screenings.

2. Biomarker Expansion & Cohorts

- Expand screening dataset to include pediatric and juvenile cohorts below 21 years of age.
- Incorporate Glycated Hemoglobin (HbA1c) and lipid profiles (Triglycerides, HDL) to enhance precision.
- Study region-specific dietary and genetic risk factors across Sudan and the Middle East.

3. Clinical Trials & Deployment

- Conduct prospective observational clinical trials in primary healthcare centers to evaluate clinician experience.
- Adopt the Lifestyle & Nutrition Agent within national preventive healthcare and diabetes awareness campaigns.
- Continuously update clinical reasoning prompts with each annual release of ADA Standards of Care.

Annex & Index

Project Deliverables & System Architecture Index

1. Official Academic Deliverables Suite

• Deliverable 1: GitHub Repository with code, pinned dependencies, .env.example, and 10/10 offline pytest suite.
• Deliverable 2: 10-Minute Video Link :
• Deliverable 3: Final Comprehensive Academic Report (COSC726_Project_Team_Health_Report.pdf).
• Comprehensive PowerPoint Presentations in Arabic and English (.pptx).

2. Source Code & Central CLI Controller

• Central CLI Launcher: run_system.py for unified execution, testing, and report compilation.
• Interactive Streamlit Web Dashboard: app.py (launch via streamlit run app.py).
• Automated 10/10 Test Suite passing offline in 4.24s with zero network dependencies.
• Calibrated Pipeline Models: calibrated_diabetes_model.pkl & clinical_preprocessor.pkl.
• Rigid Pydantic Schemas: schemas.py enforcing type safety across all 5 clinical agents.
• Deterministic ADA expert rules & hybrid multi-provider LLM connectors (Gemini, Ollama, etc.).

Thank You



Thank You for Your Kind Attention

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Al-Neelain University — Faculty of Computer Science & Information Technology
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